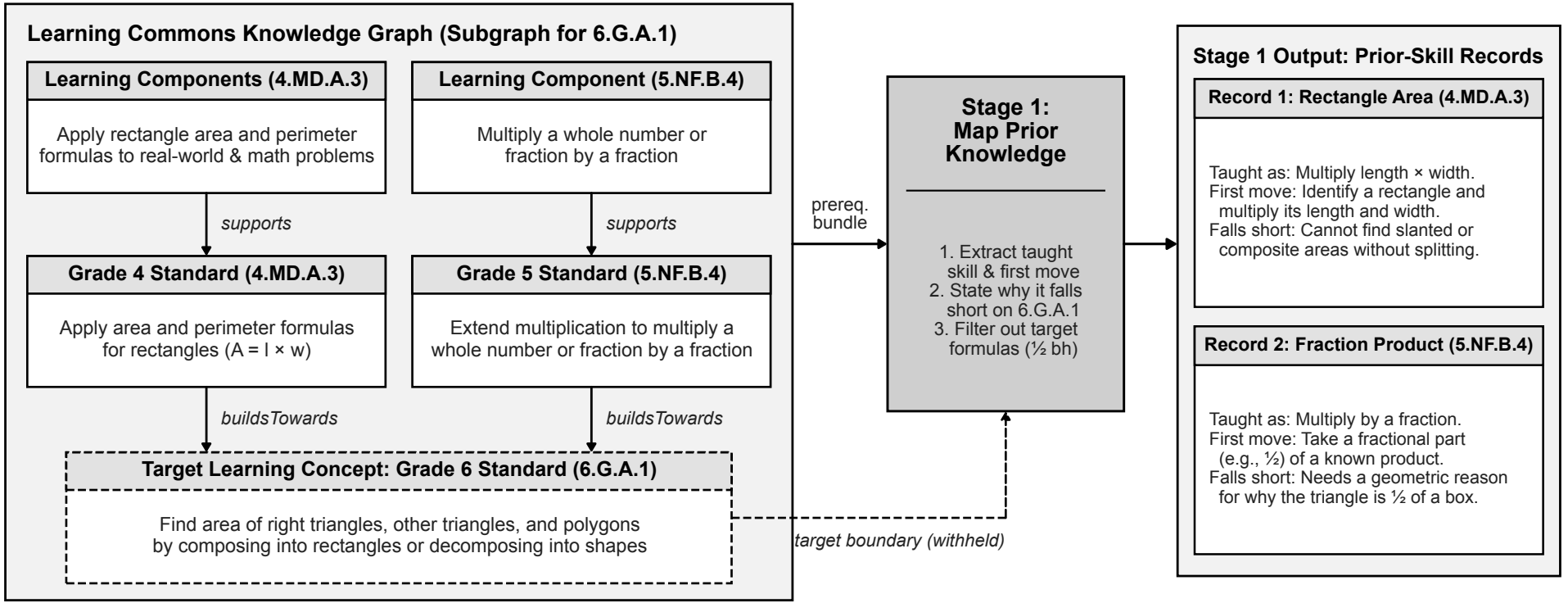

Learning Commons Knowledge Graph (Subgraph for 6.G.A.1)
Learning Components (4.MD.A.3)
Apply rectangle area and perimeter formulas to real-world & math problems
Learning Component (5.NF.B.4)
Multiply a whole number or fraction by a fraction
supports
supports
Grade 4 Standard (4.MD.A.3)
Apply area and perimeter formulas for rectangles (A = l × w)
Grade 5 Standard (5.NF.B.4)
Extend multiplication to multiply a whole number or fraction by a fraction
buildsTowards
buildsTowards
Target Learning Concept: Grade 6 Standard (6.G.A.1)
Find area of right triangles, other triangles, and polygons by composing into rectangles or decomposing into shapes
prereq. bundle
target boundary (withheld)
Stage 1: Map Prior Knowledge
1. Extract taught skill & first move
2. State why it falls short on 6.G.A.1
3. Filter out target formulas (½ bh)
Stage 1 Output: Prior-Skill Records
Record 1: Rectangle Area (4.MD.A.3)
Taught as: Multiply length × width.
First move: Identify a rectangle and multiply its length and width.
Falls short: Cannot find slanted or composite areas without splitting.
Record 2: Fraction Product (5.NF.B.4)
Taught as: Multiply by a fraction.
First move: Take a fractional part (e.g., ½) of a known product.
Falls short: Needs a geometric reason for why the triangle is ½ of a box.